# Learning Charge–Light Correlations: A Multi-Modal Foundation Model for Liquid-Argon TPCs

Paper plan and initial draft (v0)

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**How to read this draft.** Black text is meant to survive into the paper (in rough form). Red marks plan items, TODOs and placeholders. Figures and tables are PLACEHOLDERs from the tech notes, each with a note on how the final version differs. Appendix C is the campaign plan: decisions, the Section-2 job plan (≤200 node-hours), timeline and venue.

**Abstract**

Foundation models (FMs) pretrained with self-supervision are changing how detector data are analysed, but the FMs proposed for liquid-argon time projection chambers (LArTPCs) see only the ionization charge. LArTPCs also record scintillation light, which gives the event time and an independent calorimetric view. We present a multi-modal FM that reads low-level charge hits from three wire planes and the waveforms of 32 photomultipliers as one token sequence, and is pretrained without labels on MicroBooNE open data. On five tasks – $\nu$ hit identification, $\nu$/cosmic event classification, flavour, energy and charge→light prediction – light helps every model, but learning the charge–light correlation *during pretraining* (early fusion) is what matters: the same backbone pretrained on charge alone, with light added afterwards by a strong cross-attention head, recovers only part of the gain. The correlation is learned from 2D wire-plane data, without an explicit 3D reconstruction. We compare four FM recipes – a Transformer with masked autoencoding, a Transformer with a joint-embedding predictive objective, a Point Transformer V3 with self-distillation, and a Mamba-2 state-space model – with light in pretraining at matched size and compute. [**TODO:** one sentence with the Section-2 outcome.]

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*Prepared with the assistance of Claude (Anthropic). Code: `https://github.com/HaiwangYu/fm4npp-lartpc`. Tracking: `docs/paper-campaign.md` and GitHub issue #7. Every figure and number marked PLACEHOLDER comes from tech notes 1–4 (`docs/technote-round{1..4}.pdf`) and will be replaced by the campaign runs.

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# 1 Introduction

*Outline (about 1 page). References from* `docs/survey-round1.pdf`.

1. **FMs work for detector data.** Self-supervised pretraining on unlabelled detector data gives reusable representations that reduce label needs and transfer across tasks: masked particle modelling on jets [1, 2], PoLAr-MAE [3] and Panda [4, 5] on LArTPC charge, FM4NPP on a TPC [6], heterogeneous neutrino detectors [7]; review in [8].
2. **Multi-modal FMs are better where modalities are complementary.** Masked multi-modal modelling (MultiMAE [9]) and science omnimodal models (AION-1 [10]) learn cross-modal structure; in HEP, tokenized any-to-any models map between levels of the reconstruction chain [11, 12]. Native early-fusion models do at least as well as late fusion and are better at small sizes [13].
3. **The gap: LArTPC FMs are charge-only.** The scintillation light carries the event time $t_0$ (which fixes the drift coordinate) and an independent energy measure; conventional reconstruction uses it only after 3D reconstruction, through flash matching [14]. To our knowledge no LArTPC FM pretrains on charge and light together.
4. **Contributions.**
   - The first multi-modal (charge + light) self-supervised FM for LArTPCs, on low-level data (hits after signal processing, optical hits) with no high-level reconstruction.
   - A controlled study showing that the gain comes from correlations learned in pretraining, not from having light at the input (early vs late fusion; shortcut-controlled metrics).
   - A compute-matched comparison of four FM recipes with light in pretraining (Section 4).
   - Analysis of how the correlation is learned: probes, fusion depth, an oracle 3D input (Section 5).
   - A five-task benchmark on MicroBooNE open data with Pandora references. **[TODO:** decide whether to release splits/code as a benchmark.]

# 2 Data, tasks and models

**Data.** MicroBooNE BNB inclusive and $\nu_e$ CC overlay samples (NoWire HDF5 open data, Zenodo 8370883 and 7261921): simulated neutrinos on real off-beam cosmic data, about 831k events. Inputs: reconstructed wire hits of the three planes (after signal processing and hit finding) and optical hits of the 32 PMTs in a 24 $\mu$s window. No Pandora output is used as model input. Splits: 90% pretraining pool, labelled pools of 0.1%, 1%, 10%, 100%; fixed validation and test subsets (TN1).

**Tasks and metrics** (updated from the tech notes to avoid trivial learning).

**Table 1:** Downstream tasks. Changes from the tech notes in red.

| Task | Target | Headline metric |
|---|---|---|
| T1 | per-hit $\nu$ vs cosmic | per-event F1 of $\nu$ hits, unweighted median over events (up-weights small, hard events; TN4 used charge-weighted) |
| T2 | event: in-AV $\nu$ / out-of-AV $\nu$ / no $\nu$ | AUC in-AV vs no-$\nu$ on events passing the 200 PE beam-flash cut (TN4), where flash presence alone gives only AUC 0.69 |
| T3 | flavour: $\nu_\mu$ CC / $\nu_e$ CC / NC | macro-F1 (as TN1) |
| T4 | log $E_\nu$ for CC, fiducial | resolution on a truth-based clean subset, the same for every model: CC, fiducial vertex, $\nu$ carries $\geq$5% of the event charge (keeps 69%; median 7%), visible fraction $E_{\text{dep}}/E_\nu \geq 0.3$ (containment proxy); all eligible events in the appendix |
| T5 | charge $\rightarrow$ 32 PMT $\times$ 17 light features | test MSE in log(1 + PE); total-PE resolution on $\geq$200 PE events |

**Models.** Each model = tokenizer + backbone + SSL objective + downstream heads, about 20M parameters.

- **qlfm-M CL** (main): charge cells (16 wires $\times$ 64 ticks) and 32 PMT tokens in one sequence; 12-layer Transformer, $d = 384$; masked autoencoding (MAE [15]) over both modalities. **qlfm-M C**: identical, pretrained on charge only.
- **PTv3 + Sonata**: Point Transformer V3 [17] with Sonata self-distillation [18], the recipe of Panda [4], size-matched (21M). In the paper: "PTv3 (as used in Panda)"; we do not use Panda's weights.
- **Pandora** [14]: the reconstruction in the open data, used directly ($\nu$ slice, $\nu$ score) and through a BDT on 32 Pandora features (+ 8 light features).
- Section 4 adds Transformer + JEPA [16] and Mamba-2 [19] (FM4NPP recipe [6]), each with light.

**Heads.** Frozen backbone + head; single-task fine-tune (TN1–4); new: multi-task fine-tune, one backbone and five heads trained together, which is how an FM would be deployed.

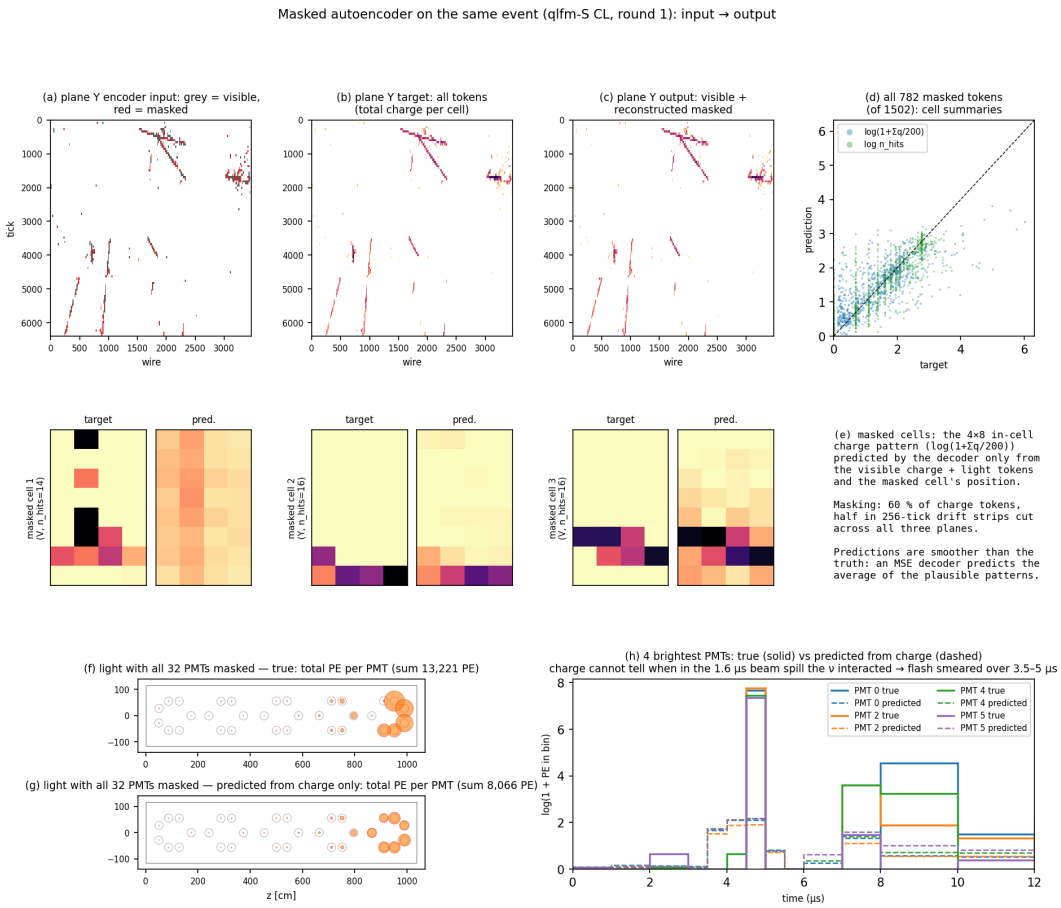



**Figure 1:** PLACEHOLDER (TN1 Fig. 5). Masked-autoencoder input and output for one event. Final: model overview – charge cells and PMT tokens → shared Transformer → MAE decoder / task heads.

# 3 Light helps, and learning the correlation is key

## 3.1 Pretraining works and light helps

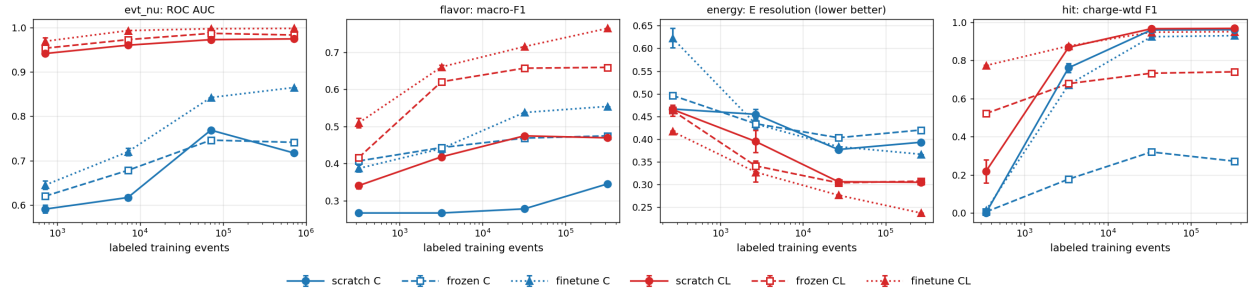



**Figure 2:** PLACEHOLDER (TN1 Fig. 6, 6.6M models). Test metrics vs labelled events; blue charge only (C), red charge + light (CL); scratch, frozen and fine-tuned. Final: qlfm-M CL vs qlfm-M C only (21M), the updated metrics of Table 1, five panels (T1–T5), frozen and multi-task fine-tune.

## 3.2 Correlations from pretraining beat light added afterwards

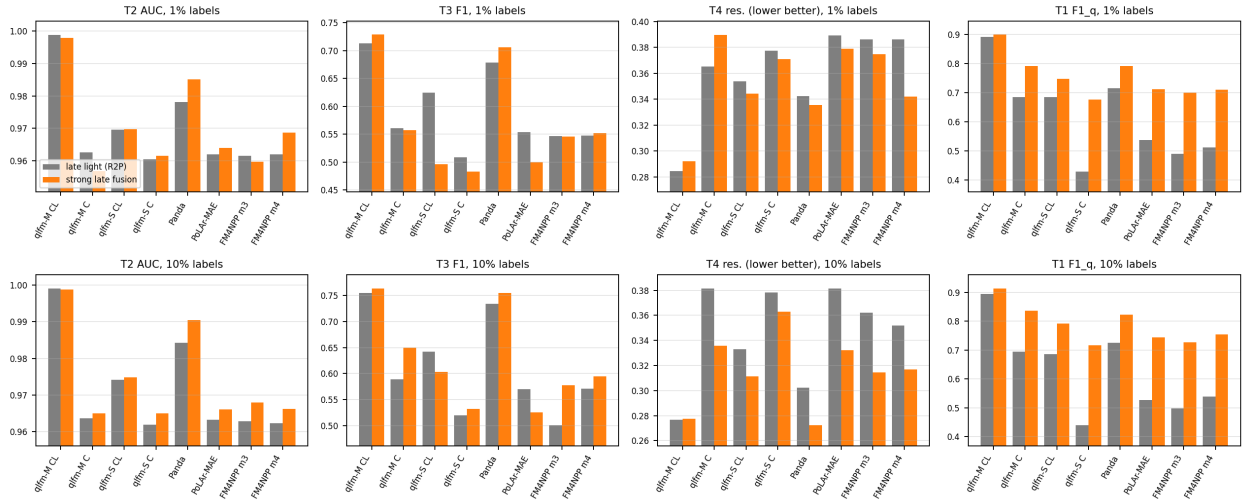



**Figure 3:** PLACEHOLDER (TN3). Frozen backbones at 1% (top) and 10% labels (bottom) with weak late light (grey) and strong late fusion (orange, cross-attention from charge to PMT tokens). qlfm-M CL has light in pretraining; all other backbones are charge-only. Final: keep, restrict to qlfm-M CL, qlfm-M C, PTv3 + Sonata; updated metrics.

**Table 2:** PLACEHOLDER values from TN2–4 (frozen, 1% labels, seed 0; T1 charge-weighted median per-event F1; T2 on the 200 PE pass sample; T4 on all CC fiducial events). Late fusion = strong cross-attention head. Final: 0.1% and 1% labels, frozen and multi-task fine-tune, updated metrics, 3 seeds; C-only diagnostic rows (light removed at inference).

| Model | Light | T1 F1 ↑ | T2 AUC ↑ | T3 F1 ↑ | T4 res. ↓ | T5 MSE ↓ |
|---|---|---|---|---|---|---|
| qlfm-M CL | pretraining | **0.91** | **0.99** | **0.713** | **0.284** | **0.503** |
| qlfm-M C | late head | 0.80 | 0.82–0.86 | 0.558 | 0.390 | 0.586 |
| PTv3 + Sonata | late head | 0.79 | 0.95–0.97 | 0.706 | 0.336 | 0.577 |
| Pandora (+ BDT) | BDT features | 0.81 | 0.95 | 0.559 | 0.400 | n/a |

## 3.3 Shortcut control and charge→light prediction

The large-light cut (T2) and the charge-only T5 task test whether the model uses charge–light consistency rather than the presence of a flash. qlfm-M CL predicts the PMT light from charge alone, and its own MAE decoder does so with no labels at all (MSE 0.506 vs 0.503 with a trained head; charge-only encoders 0.58–0.59).

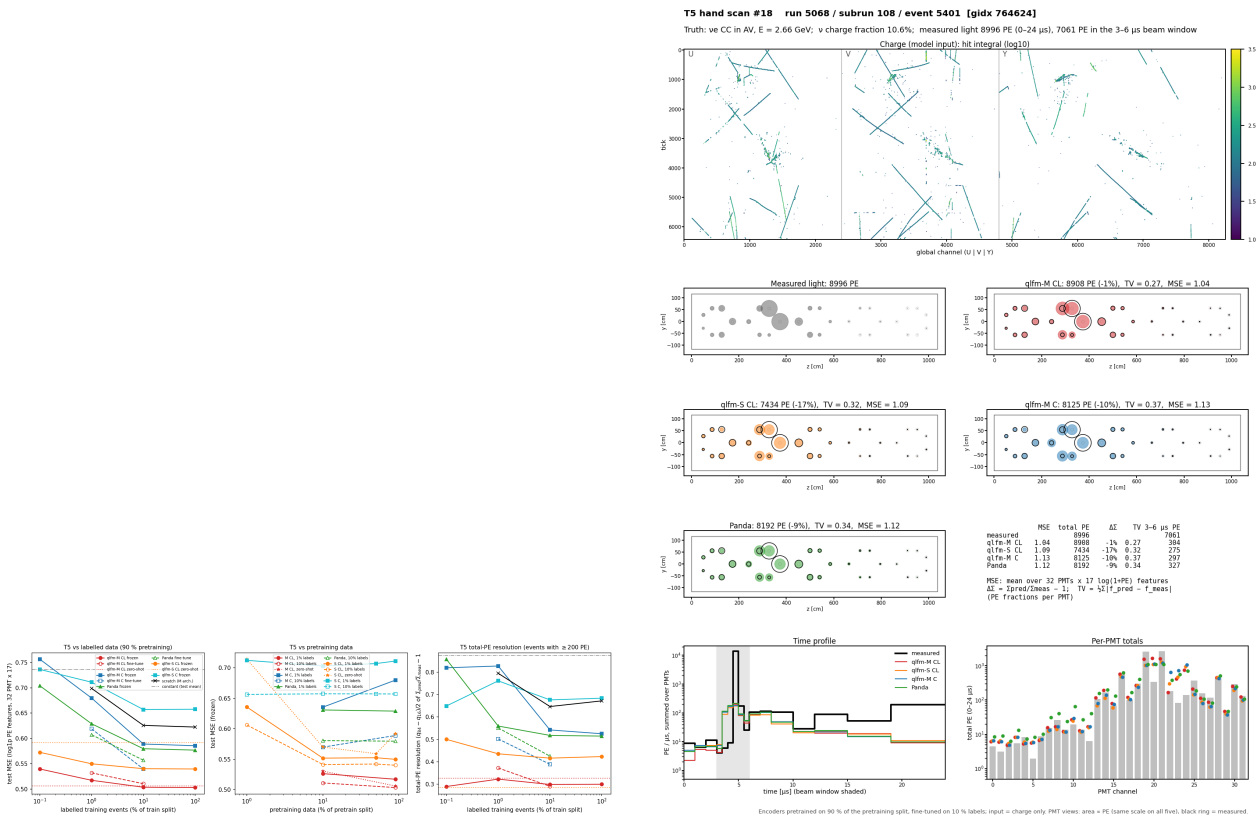



**Figure 4:** PLACEHOLDER (TN4). Left: T5 test MSE vs pretraining and label fraction. Right: one hand-scan event (measured vs predicted PMT light). Final: T5 bar chart for the four models + zero-shot, and one display.

### 3.4 Multi-task fine-tuning

[**TODO:** New. One backbone, five heads; loss per task masked to its eligible events, weights normalized by each task's initial loss. Compare with single-task fine-tunes at 0.1% and 1%: how much does sharing the backbone cost?]

**Table 3:** PLACEHOLDER layout. Single-task (ST) vs multi-task (MT) fine-tune, qlfm-M CL. Values pending.

| Labels | Fine-tune | T1 | T2 | T3 | T4 | T5 |
|---|---|---|---|---|---|---|
| 0.1% | ST / MT | – / – | – / – | – / – | – / – | – / – |
| 1% | ST / MT | – / – | – / – | – / – | – / – | – / – |

## 4 Which multi-modal FM architecture suits LArTPCs?

So far only qlfm saw light in pretraining; the references got light only at the head. Other recipes may absorb the charge–light correlation better. We give each recipe light in pretraining and compare at matched size (≈20M) and compute (20 node-hours of pretraining).

**Table 4:** The four recipes. Architecture and objective change together except for the MAE/JEPA pair, which isolates the objective on one backbone.

| Recipe | Backbone | Objective | How light enters pretraining |
|---|---|---|---|
| qlfm-M CL | Transformer | MAE (charge + light) | PMT tokens in the sequence (as now) |
| qlfm-M CL JEPA | Transformer | JEPA (I-JEPA-style) | same tokens; predictor regresses EMA-target embeddings of masked charge blocks and masked PMTs |
| PTv3 + Sonata + L | PTv3 | self-distillation | event-level cloud: 3 planes plus 32 PMT points in a fourth slab, placed at their $(z, y)$; per-modality input embedding [**TODO:** verify in smoke test] |
| Mamba-2 + L | Mamba-2 (FM4NPP m4) | k-NNN + masked light | PMT tokens as prefix (conditioning) and suffix (masked light regressed from the charge); hits as in FM4NPP |

Convergence: validation loss vs epoch for each recipe (own objective), and a frozen probe (T2 bright AUC, T5 MSE) at 2, 5, 10, 20 node-hours – the only curve comparable across objectives.

**Figure 5:** PLACEHOLDER Pending (Appendix C).

**Table 5:** PLACEHOLDER layout. Multi-task fine-tune at 1% labels (selection metric), and frozen at 1%; 3 seeds.

| Recipe | T1 | T2 | T3 | T4 | T5 | mean rank |
|---|---|---|---|---|---|---|
| qlfm-M CL (MAE) | – | – | – | – | – | – |
| qlfm-M CL (JEPA) | – | – | – | – | – | – |
| PTv3 + Sonata + L | – | – | – | – | – | – |
| Mamba-2 + L | – | – | – | – | – | – |

# 5 How the correlations are learned

- **Embedding maps.** t-SNE / PCA of frozen event and token embeddings without a learned map, coloured by $\nu$/cosmic, flavour, energy, drift coordinate (as FM4NPP Fig. 8 [6]).
- **Probes.** Linear probes on charge tokens only (light masked at inference) for the drift coordinate $x$ (which needs $t_0$), deposited energy and the vertex: CL vs C encoder.
- **Fusion depth** (survey E1). Light enters at the input, after $L/2$ blocks, or only at the head, all at matched compute.
- **Oracle 3D input** (appendix-level). The open data has no space points; a 3D input can only be truth energy deposits. It gives an upper bound on what 3D reconstruction could add, and is labelled as an oracle.

- **Light-swap counterfactual** (survey E2, optional): replace an event's light by another event's and measure the drop.
- **Validity checks** (Josh Peterson's survey round 1.5, after Aobo Li's papers; inference only, <1 nh each): shuffled-label and plane/PMT-group ablations on T1/T2 (A2); MAE reconstruction loss vs T2 score for $\nu$ and cosmic events, to rule out an internal shortcut (A5); T1 efficiency vs drift coordinate, wire plane and hit amplitude (A3); data vs simulation: T5 errors on no-$\nu$ events (real off-beam cosmic light) vs $\nu$ events (simulated light) (A1, partial: no beam-off sample in the open data).

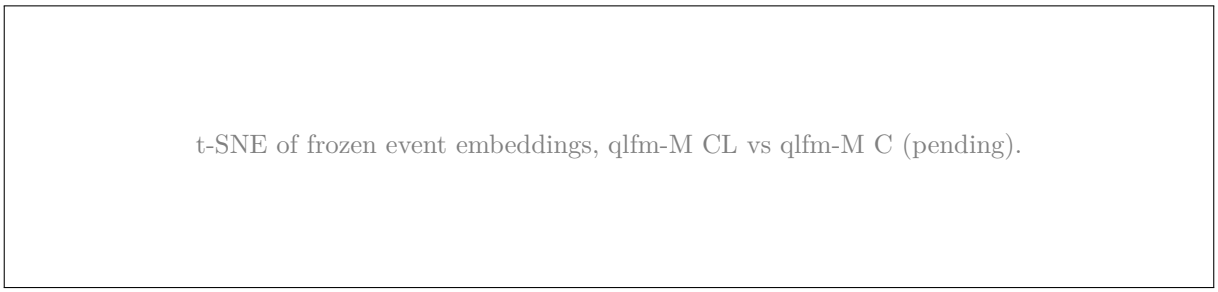



**Figure 6:** PLACEHOLDER Pending.

## 6 Scaling

On hold until Sections 3–5 are done and the go-ahead is given. Scaling of each recipe with model size and compute; smaller charge cells (more tokens), where SSMs may gain (Josh's SSM survey).

## 7 Discussion and limitations

Overlay simulation (no beam-on data); one detector; models of about 20M parameters; compute-limited training; T4 on a clean subset. [**TODO:** expand.]

## A Model setups

[**TODO:** One subsection per recipe: tokenizer, backbone, objective, hyper-parameters, pretraining compute, heads; MAE input/output and token figures from TN1 (Figs. 2–4).]

# B Event displays

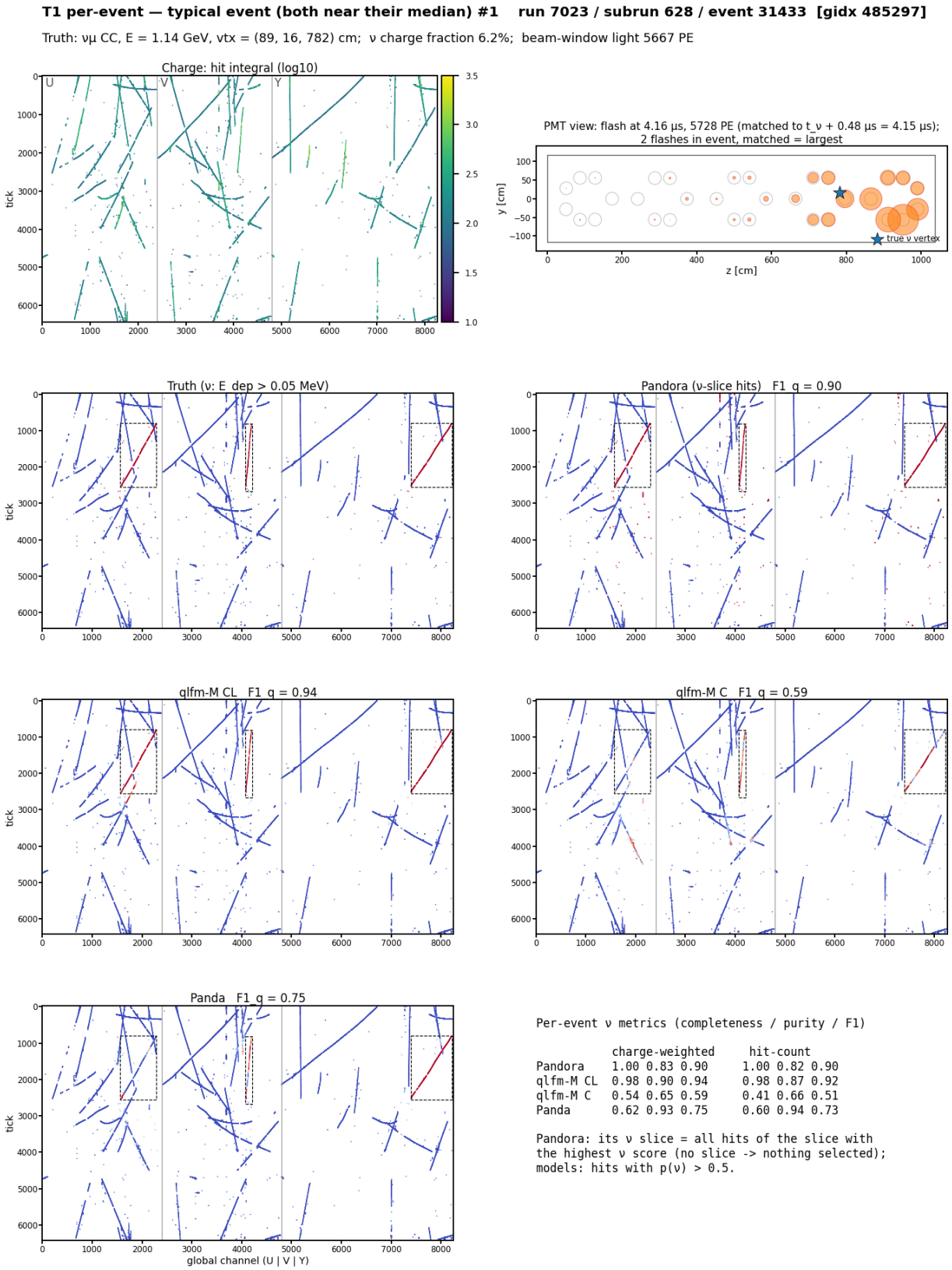



**Figure 7:** PLACEHOLDER (TN4). A typical event: T1 hit predictions of qlfm-M CL, qlfm-M C, PTv3 and Pandora.

# C Campaign plan

## C.1 Decisions so far (6 Nov 2026)

- Section 4 first; total budget ≤200 node-hours (nh), charged to `neutrinoGPU::WireCellDev`, preemptable queue.
- All four recipes pretrained at the same budget, 20 nh each, including a new qlfm-M CL run (the existing one used about 2 nh, the round-2 references about 4 nh).
- T4 scored on a truth-based subset common to all models (not on each model's T1): $\nu$ share of the event charge ≥5%, visible fraction ≥0.3.
- Benchmark release (splits, tasks, metrics, baselines). Josh Peterson's code audit is fixed before any campaign run; his checks A1–A5 go into Section 5.
- JEPA on the qlfm Transformer, so MAE vs JEPA isolates the objective.
- "Reco-3D" becomes an oracle study with truth deposits (no space points in the open data).
- Naming: PTv3 + Sonata ("as used in Panda"), not "Panda" and not "PT2".
- Venue: ICML 2027 main conference, *Application-Driven ML* area (deadline expected late January 2027); fallbacks: an ICML 2027 workshop, the NeurIPS 2027 Datasets & Benchmarks track, MLST.

## C.2 Section 4 job plan

**Step 0 – code (login node + short debug jobs).**

1. Multi-task fine-tune mode in `train_downstream` (five heads, masked per-task losses, loss normalization); works for all backbones through the common interface.
2. Light for PTv3: event-level clouds (3 planes + PMT points), Sonata views include PMTs; pretraining config `ptv3_sonata_cl`.
3. Light for Mamba-2: PMT prefix and suffix tokens, masked-light regression added to the k-NNN loss.
4. JEPA objective on the qlfm backbone (context encoder, EMA target encoder, narrow predictor; block masks on each plane plus masked PMTs); collapse monitor (embedding rank).
5. Logging: validation loss every ≈15 minutes; checkpoints kept at 2, 5, 10, 20 nh for the probe curve.

qlfm-M CL needs no new code and starts first.

**Step 1 – smoke tests.** 30-minute runs of each new recipe: loss falls, no NaN, throughput measured (sets the step count for 20 nh), resume after a kill works.

**Step 2 – pretraining.** Four 1-node preemptable jobs of 20 h each (`-r y`, resumable), running in parallel; the feeder keeps at most 6 jobs in the queue.

**Step 3 – downstream.** For each recipe, using the 1-node `queue.pbs` worker (4 tasks per node):

- frozen heads, T1–T5, at 0.1% and 1%;
- multi-task fine-tune at 0.1% and 1%, 3 seeds (selection metric: 1%);
- single-task fine-tunes of qlfm-M CL at 1% (for Table 3);
- probe curve: frozen T2 and T5 at 1% on the 2/5/10/20 nh checkpoints.

**Table 6:** Budget for Section 4 (node-hours). Downstream costs from rounds 1–4: a frozen or single-task head run takes 0.1–0.3 nh per GPU slot (4 per node); PTv3 fine-tunes are about 2× slower.

| Item | per recipe | total | note |
|---|---|---|---|
| Code and smoke tests | 3–4 | 13 | debug queue |
| Pretraining | 20 | 80 | preemptable, 1 node × 20 h |
| Probe curve (4 ckpts, T2+T5) | 1.5 | 6 | |
| Frozen heads (5 tasks, 2 fractions) | 1.5 | 6 | |
| Multi-task FT (2 fractions, 3 seeds) | 3–5 | 16 | |
| Single-task FT (qlfm-M CL, 1%) | – | 2 | for Table 3 |
| Reruns, preemption (about 25%) | – | 30 | |
| **Total** | | **≈155** | cap 200 (45 spare) |

The 45 nh spare can pay for charge-only twins of the new recipes or for the fusion-depth ablation of Section 5.

### C.3 Timeline (target)

| | |
|---|---|
| Week of 9 Nov | Step 0–1; qlfm-M CL 20 nh pretraining starts |
| 16–29 Nov | PTv3 + L, Mamba-2 + L, JEPA pretraining |
| 30 Nov – 13 Dec | downstream, probe curves; Section 4 results |
| 14 Dec – 10 Jan | Section 3 updates (metrics, multi-task FT), Section 5 |
| 11–27 Jan | writing; ICML deadline (expected late January) |

### C.4 Risks

- **PTv3 + L design** (PMT points in a point-cloud backbone) and Sonata's view augmentations may not suit PMT points; fall back to a light cross-attention adapter inside the encoder stages (mid fusion) and say so.
- **JEPA collapse**: our earlier self-distillation on qlfm tokens (qlfm-SD) underperformed, probably due to a low learning rate; monitor embedding rank, tune the LR in the smoke test.
- **Mamba-2 and masking**: SSMs fit MAE poorly (Josh's SSM survey); we keep FM4NPP's k-NNN objective and add masked light at the suffix.
- **Preemption**: runs resume from checkpoints; validated only with a normal job so far.

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